

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			Indicative content: 1. Set up a pointer in the clamp stand and adjust its height. 2. The height (distance 2) should be at least 100 cm above the floor. Measure and record this height using a metre ruler. 3. Drop a single cake case from a height (distance 1) at least 15 cm above the pointer. 4. Using a stopwatch, measure and record the time it takes to fall from the pointer to the floor. 5. Repeat steps 3 and 4. 6. Repeat by adding more cake cases to the stack. 5–6 marks Detailed description of apparatus set-up and suitable values and the measurements taken. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3–4 marks Detailed description of either apparatus set-up and suitable values or the measurements taken OR limited description of both. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i>	6			6		6

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				1–2 marks Limited description of either apparatus set-up and suitable values or the measurements taken. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)	(i)		1.5		1		1	1	1
		(ii)	I	4 points correct < 1 small square (2) not including (0,0) and (1.5, 4.8) 3 points correct < 1 small square (1) 2 points or less correct (0)		2		2	2	2
			II	(1.5,4.8) not judge the accuracy of the point		1		1		1
			III	Reasonable best fit curve starts at the origin		1		1	1	1
		(iii)		Read from candidate’s graph for a mass of 0.5 [g] Allow ± 0.1 m/s tolerance		1		1	1	1
				Question 4 total	6	6	0	12	5	12